

90-Day C Programming Mastery Plan

◆ Level 1 — Basics (Days 1–10)

Goal: Be comfortable with input/output and conditions.

Programs:

Hello World

Add two numbers

Swap numbers (with temp)

Swap numbers (without temp)

Odd / Even number

Largest of two numbers

Largest of three numbers

Positive / Negative / Zero

Simple calculator (if-else)

Menu-driven calculator (switch)

◆ Level 2 — Loop Logic (Days 11–25)

Goal: Master for, while, do-while.

Programs: 11. Print numbers 1–100

12. Sum of 1–N numbers

13. Factorial

14. Reverse number

15. Sum of digits

16. Count digits
17. Palindrome number
18. Armstrong number
19. Multiplication table
20. Fibonacci series
21. Prime number check
22. Prime numbers between 1–N
23. Perfect number
24. Strong number
25. Sum of even & odd numbers

◆ Level 3 — Pattern Thinking (Days 26–40)

Goal: Improve loop thinking and logic.

Programs: 26. Half star pyramid

27. Inverted pyramid
28. Full pyramid
29. Diamond pattern
30. Number pyramid
31. Floyd's triangle
32. Pascal triangle (basic)
33. Alphabet triangle
34. Reverse number triangle
35. Continuous number pattern

36. Hollow square

37. Hollow pyramid

38. Increasing number pattern

39. Decreasing number pattern

40. Cross pattern

◆ Level 4 — Arrays (Days 41–60)

Goal: Start thinking like a real programmer.

Programs: 41. Input and display array

42. Sum of array elements

43. Largest element in array

44. Smallest element in array

45. Average of array

46. Reverse array

47. Count even/odd elements

48. Linear search

49. Binary search

50. Sort array ascending

51. Sort array descending

52. Merge arrays

53. Find duplicates

54. Remove duplicates

55. Frequency of elements

56. Second largest element

57. Rotate array left

58. Rotate array right

59. Matrix addition

60. Matrix multiplication

◆ Level 5 — Strings (Days 61–75)

Goal: Work with text data.

Programs: 61. Input and display string

62. String length

63. Reverse string

64. Palindrome string

65. Count vowels and consonants

66. Copy string

67. Concatenate strings

68. Compare strings

69. Convert lowercase → uppercase

70. Count words in sentence

71. Remove spaces

72. Frequency of characters

73. Check anagram

74. Find substring

75. Replace character

♦ Level 6 — Functions (Days 76–85)

Goal: Write structured programs.

Programs: 76. Function to add numbers

77. Function for factorial

78. Function for prime check

79. Function for Fibonacci

80. Function for palindrome

81. Function returning largest number

82. Recursive factorial

83. Recursive Fibonacci

84. Function for GCD

85. Function for LCM

♦ Level 7 — Mini Projects (Days 86–90)

Goal: Combine everything.

Projects: 86. Student marks system

87. Simple bank system

88. Quiz program

89. File write + read program

90. Mini project using arrays + functions